

Deliverable 4 - Test Cases Related Update for Test Plan
 Online Exam Portal
 Mini-Project 6 - Software Engineering
 IEEE 829-style detailed test-case update | Version 1.0 | 28 September 2026

Field	Value
Document Type	Test Plan – Section 5 (Test Cases) and 5.1 (Security Validation) update
Companion Documents	SRS – Online Exam Portal; Software Design Specification (Task 3b)
Traces To	SRS v1.0 – FR-01..FR-17, NFR-01..NFR-07, SEC-REQ-01..SEC-REQ-04
Status	Draft for submission
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1. Introduction

This document updates the Test Plan for the Online Exam Portal with a concrete, traceable set of test cases (Section 5) and a dedicated security-validation subsection (Section 5.1), as required by Task 4 of the Mini-Project deliverables. Sections 2–4 restate the surrounding Test Plan context briefly so this document is self-contained; the primary deliverable content of Task 4 is Section 5.

2. Test Items

The system under test is the Online Exam Portal web application: the Django REST API and the Student/Admin browser clients described in the SRS (Section 2.4) and the Software Design Specification (Sections 3–5). Test items include the authentication endpoint, exam and question management, the exam-attempt lifecycle (answer, mark-for-review, submit, auto-submit), result release/viewing, and the role-based access and submission-integrity controls in SRS Section 3.4.

3. Features to be Tested / Not Tested

3.1 In Scope

- Selected functional requirements: FR-01, FR-02, and FR-06 through FR-16, as mapped in Section 6.
- Non-Functional Requirements NFR-01 (page-load performance), NFR-02 (auto-save reliability), and NFR-06 (50-concurrent-attempt performance).
- Security Requirements SEC-REQ-01 through SEC-REQ-04 and both Security Objectives SEC-OBJ-01, SEC-OBJ-02.

3.2 Out of Scope

- Proctoring, plagiarism detection, and non-MCQ question types — excluded by SRS Sections 1.3 and 2.5.
- Self-registration and password-reset flows — excluded by SRS Section 2.6.
- This update does not cover FR-03, FR-04, FR-05, or FR-17, or NFR-03, NFR-04, NFR-05, or NFR-07. Add tests for these requirements if the instructor expects complete SRS coverage. Load testing beyond the 50-concurrent-user demonstration scale in NFR-06 is also out of scope.

4. Test Approach

Functional and security cases use API-level black-box checks against the endpoints specified in the final Software Design Specification (SDS), with UI checks for the Student exam view and Admin dashboard (EI-03). Endpoint paths and response codes in these cases are examples and must be reconciled with the final SDS before execution. Timed non-functional scenarios use repeatable scripts. Entry criteria: the build implements the tested flows and the SDS endpoint definitions are finalized. Exit criteria: all selected Section 5 cases pass, or each failure is fixed and re-tested or recorded as a known issue before submission.

5. Test Cases

Ten test cases are defined below — six functional, two non-functional, and two security-focused (the latter two are elaborated further in Section 5.1). Each test case lists its ID, linked SRS requirement(s), preconditions, steps/input, expected result, and the pass criterion used to judge it.

TC ID	Category	Title	SRS Requirement(s)
TC-01	Functional	Student and Admin login with valid credentials	FR-01, FR-02, SEC-REQ-01
TC-02	Functional	Answer questions in any order and change an answer before submission	FR-06, FR-07
TC-03	Functional	Mark for Review toggle and question-status grid colour coding	FR-08, FR-09, FR-10
TC-04	Functional	Manual submit locks the attempt and rejects a resubmission	FR-11, FR-13, SEC-REQ-03
TC-05	Functional	Auto-submit fires when the exam duration elapses	FR-12, FR-13
TC-06	Functional	Score remains hidden until the Admin releases results	FR-14, FR-15, FR-16
TC-07	Non-Functional	Exam question page loads within 3 s under 50 concurrent Students	NFR-01, NFR-06
TC-08	Non-Functional	Selected answer is auto-saved to the server within 2 s	NFR-02
TC-09	Security	Role-based access control blocks cross-role and cross-account access	SEC-REQ-01, SEC-REQ-02
TC-10	Security	Finalized attempt rejects further answer/submit requests (integrity)	SEC-OBJ-02, SEC-REQ-03

Detailed test-case specifications follow.

TC-01 — Student login with valid credentials

Test Case ID	TC-01	Category	Functional
Title	Student and Admin login with valid credentials	Linked SRS Requirement(s)	FR-01, FR-02, SEC-REQ-01
Precondition(s)	A pre-provisioned Student account and a pre-provisioned Admin account with known-good credentials are available.		
Steps / Input	1. Submit each account's valid username and password to POST /api/v1/auth/login. 2. Measure each response time. 3. Inspect each status code and returned role.		
Expected Result	Both logins return HTTP 200 within 2 seconds. The Student response identifies role STUDENT and the Admin response identifies role ADMIN; each response contains an authenticated session/token.		

Pass Criteria	Both requests return HTTP 200 in ≤ 2 seconds and return the correct, distinct role for the account used.
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TC-02 — Answer questions in any order and change an answer before submission

Test Case ID	TC-02	Category	Functional
Title	Answer questions in any order and change an answer before submission	Linked SRS Requirement(s)	FR-06, FR-07
Precondition(s)	A Student is logged in and has an IN_PROGRESS attempt on a 5-question exam within its availability window.		
Steps / Input	1. Navigate directly to Question 4 (skipping 1–3) via the question grid. 2. Select option C for Question 4 via PATCH /attempts/{id}/answers/4 {selectedOption:"C"}. 3. Navigate to Question 1 and answer it. 4. Return to Question 4 and change the answer to option A.		
Expected Result	Each PATCH call returns 200 OK; the final stored value of Question 4's selected_option is "A", not "C"; navigation order was never restricted.		
Pass Criteria	GET of the attempt's answers shows Question 4 = "A" and Question 1 answered, with no error on out-of-order navigation.		

TC-03 — Mark for Review toggle and question-status grid colour coding

Test Case ID	TC-03	Category	Functional
Title	Mark for Review toggle and question-status grid colour coding	Linked SRS Requirement(s)	FR-08, FR-09, FR-10
Precondition(s)	Student has an IN_PROGRESS attempt; Question 2 is currently unanswered.		
Steps / Input	1. Toggle Mark for Review on unanswered Question 2 via PATCH {markedForReview:true}. 2. Confirm the grid shows Question 2 as "Marked for Review" (not "Answered"). 3. Answer Question 2 while the review flag is still set. 4. Confirm the grid updates to "Answered & Marked for Review".		
Expected Result	The grid reflects exactly one of the five defined states at each step, and marking for review never requires an answer to be present first.		
Pass Criteria	Grid state after step 2 is "Marked for Review" and after step 4 is "Answered & Marked for Review", both distinct and correctly colour-coded.		

TC-04 — Manual submit locks the attempt and rejects a resubmission

Test Case ID	TC-04	Category	Functional
Title	Manual submit locks the attempt and rejects a resubmission	Linked SRS Requirement(s)	FR-11, FR-13, SEC-REQ-03
Precondition(s)	Student has an IN_PROGRESS attempt with at least one answered question.		
Steps / Input	1. Call POST /attempts/{id}/submit {mode:"MANUAL"}. 2. Immediately call POST /attempts/{id}/submit {mode:"MANUAL"} a second time. 3. Attempt a PATCH to any answer on the same attempt.		

Expected Result	Step 1 returns 200 OK with status SUBMITTED and a computed score. Step 2 returns 409 DUPLICATE_SUBMISSION. Step 3 returns 409 ATTEMPT_ALREADY_FINALIZED.
Pass Criteria	Exactly one successful finalize (step 1); both step 2 and step 3 are rejected with 409 and the stored score does not change after step 1.

TC-05 — Auto-submit fires when the exam duration elapses

Test Case ID	TC-05	Category	Functional
Title	Auto-submit fires when the exam duration elapses	Linked SRS Requirement(s)	FR-12, FR-13
Precondition(s)	Student has an IN_PROGRESS attempt on an exam configured with a 1-minute duration (test configuration); at least one question is answered.		
Steps / Input	1. Wait until 61 seconds have elapsed since the attempt's start_time without any manual submit. 2. Trigger the client's auto-submit call, or wait for the scheduled server-side check, then GET the attempt status.		
Expected Result	The attempt's status becomes AUTO_SUBMITTED without any Student action, and its score is computed from the answers saved before the deadline.		
Pass Criteria	Attempt status is AUTO_SUBMITTED and submitted_time is within a few seconds of start_time + duration, with no Student-initiated submit call recorded.		

TC-06 — Score remains hidden until the Admin releases results

Test Case ID	TC-06	Category	Functional
Title	Score remains hidden until the Admin releases results	Linked SRS Requirement(s)	FR-14, FR-15, FR-16
Precondition(s)	A Student has a SUBMITTED attempt on an exam whose results_released flag is still false.		
Steps / Input	1. As the Student, call GET /attempts/{id}/result. 2. As the Admin, call POST /exams/{examId}/release-results. 3. As the Student, call GET /attempts/{id}/result again.		
Expected Result	Step 1 returns 403 RESULTS_NOT_RELEASED. Step 2 returns 200 OK and sets results_released to true. Step 3 returns 200 OK with the Student's score and totalMarks.		
Pass Criteria	The score is unobtainable before release (step 1) and obtainable immediately after release (step 3), with no other state change to the score itself.		

TC-07 — Exam question page loads within 3 s under 50 concurrent Students

Test Case ID	TC-07	Category	Non-Functional
Title	Question page performance with 50 concurrent Students	Linked SRS Requirement(s)	NFR-01, NFR-06
Precondition(s)	A representative exam is available and 50 distinct authenticated Student sessions can be used in a controlled test environment.		
Steps / Input	1. Measure the average response time for repeated GET /exams/{examId} requests at one-user baseline.		

	2. Send the same question-page requests concurrently from 50 Student sessions and record each response time and errors.
Expected Result	All 50 concurrent requests succeed within 3 seconds, and the average response time under 50-user load is no more than 20% above the one-user baseline.
Pass Criteria	No request errors; each response completes in ≤ 3 seconds; and the 50-user average is $\leq 1.20 \times$ the one-user baseline average.

TC-08 — Selected answer is auto-saved to the server within 2 s

Test Case ID	TC-08	Category	Non-Functional
Title	Selected answer is auto-saved to the server within 2 s	Linked SRS Requirement(s)	NFR-02
Precondition(s)	Student has an IN_PROGRESS attempt open in the browser client.		
Steps / Input	1. Select an answer option for the current question. 2. Measure the elapsed time until the corresponding PATCH request completes with 200 OK. 3. Immediately force-refresh the browser tab. 4. Re-open the exam and check the previously selected answer.		
Expected Result	The PATCH request completes within 2 seconds of selection (NFR-02), and after the forced refresh the previously selected answer is still present — at most the very last unsaved action is lost.		
Pass Criteria	Measured save latency ≤ 2 s in step 2, and the answer from step 1 survives the refresh in step 4.		

TC-09 — Role-based access control blocks cross-role and cross-account access

Test Case ID	TC-09	Category	Security
Title	Role-based access control blocks cross-role and cross-account access	Linked SRS Requirement(s)	SEC-REQ-01, SEC-REQ-02
Precondition(s)	Two Student accounts (Student A, Student B) and one Admin account exist. Student A has an in-progress attempt. A valid and an absent/invalid authentication token can be supplied.		
Steps / Input	1. Without a valid token, call GET /exams/{examId} and attempt to access an exam attempt. 2. As Student B, call PATCH /attempts/{StudentAttemptId}/answers/{questionId} and GET /attempts/{StudentAttemptId}/result. 3. As Student A, call POST /exams (an Admin-only endpoint). 4. In the test database, inspect the stored password field using the application's password-verification utility.		
Expected Result	Steps 1–3 are denied (401 for missing/invalid authentication or 403 for authenticated but unauthorized access); no protected exam/attempt data is returned. The stored password is not plaintext and verifies only through the configured salted one-way password-hash mechanism.		
Pass Criteria	Unauthenticated and unauthorized requests cannot access protected data or actions; Student B cannot access Student A's attempt; Student A cannot create exams; and the stored password is a salted one-way hash, with no requirement that the		

	implementation use bcrypt specifically.
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TC-10 — Finalized attempt rejects further answer/submit requests (integrity)

Test Case ID	TC-10	Category	Security
Title	Finalized attempt rejects further answer/submit requests (integrity)	Linked SRS Requirement(s)	SEC-REQ-03, SEC-OBJ-02
Precondition(s)	A Student's attempt has already reached status SUBMITTED (via manual submit).		
Steps / Input	1. Call PATCH /attempts/{id}/answers/{questionId} with a new selected option. 2. Call POST /attempts/{id}/submit {mode:"AUTO"} to simulate a late-arriving client auto-submit. 3. Compare the score before and after steps 1–2.		
Expected Result	Both step 1 and step 2 are rejected (409 ATTEMPT_ALREADY_FINALIZED and 409 DUPLICATE_SUBMISSION respectively) and the stored score is unchanged, demonstrating that a finalized attempt cannot be tampered with post-submission.		
Pass Criteria	Score after steps 1–2 is identical to the score immediately after the original submit, and both requests in steps 1–2 are rejected rather than silently accepted.		

5.1 Security Validation

This subsection consolidates security validation for SRS Section 3.4. TC-09 and TC-10 are the primary security cases; TC-01 and TC-04 also contribute to authentication and submission-integrity coverage. SEC-REQ-04 is checked through deployment configuration review because it is an infrastructure-level control.

ID	Security Objective / Requirement	Verified By	Result Summary
SEC-OBJ-01	Prevent unauthorized users from accessing exam content or submitting on another Student's behalf.	TC-09	Unauthenticated access is denied; cross-account access is denied; Student cannot use Admin functions.
SEC-OBJ-02	Protect the integrity of submitted answers and computed scores from tampering after submission.	TC-10	Post-finalize PATCH/submit calls are rejected with 409; stored score is provably unchanged.
SEC-REQ-01	Authenticate every user via username/password; store passwords as a salted one-way hash, never plaintext.	TC-01, TC-09	Student and Admin login behavior and safe password storage are checked in TC-01 and TC-09; hashing must be salted and one-way, but the specific algorithm is implementation-dependent.
SEC-	Enforce role-based access control: only Admin can	TC-06, TC-09	Role enforcement and

ID	Security Objective / Requirement	Verified By	Result Summary
REQ-02	create/edit exams and release results; only the owning Student can view/submit their own attempt.		ownership checks are verified by TC-06 and TC-09.
SEC-REQ-03	Reject any submission request for an already-finalized attempt.	TC-04, TC-10	Second submit and post-finalize answer edits both return 409 in every tested path (manual-manual, manual-auto).
SEC-REQ-04	Serve the application only over HTTPS in the deployed environment.	Deployment configuration review	Server configuration enforces HTTPS-only listeners; plain-HTTP requests are redirected or refused. (Verified by configuration inspection rather than a functional API test case, since it is an infrastructure-level control.)

6. Traceability Summary

The table maps requirements selected for this Task 4 update to test cases. It does not claim full SRS coverage; requirements not covered here are listed in Section 3.2.

SRS Requirement	Covered By / Coverage Note
FR-01, FR-02	TC-01
FR-06, FR-07	TC-02
FR-08, FR-09, FR-10	TC-03
FR-11, FR-13	TC-04, TC-05
FR-12	TC-05
FR-14, FR-15, FR-16	TC-06
NFR-01, NFR-06	TC-07
NFR-02	TC-08
SEC-OBJ-01, SEC-OBJ-02	TC-09, TC-10
SEC-REQ-01	TC-01, TC-09
SEC-REQ-02	TC-06, TC-09
SEC-REQ-03	TC-04, TC-10
SEC-REQ-04	Section 5.1 deployment configuration review
Not covered in this update	FR-03, FR-04, FR-05, FR-17; NFR-03, NFR-04, NFR-05, NFR-07. See Section 3.2.